

The measured transition and the real one: a review of corporate decarbonisation indicators and what they predict

Working paper / review article Prepared from a corpus of fifteen studies published between 2023 and 2026.

Abstract

Corporate climate commitments have become close to universal among large listed companies, and corporate carbon reporting has improved enough that portfolio-level emissions accounting is now routine. Physical abatement has not kept pace. This review brings together fifteen recent empirical studies to argue a specific point: most of the measured improvement in corporate carbon performance sits in the parts of the accounting system that respond to procurement and disclosure decisions, and comparatively little of it sits in the parts that respond to production. Three findings recur across studies that use different data and different identification strategies. Emissions-reduction effects attributed to target-setting appear in market-based Scope 2 and disappear in location-based Scope 2. Headline declines at portfolio level are dominated by index composition and by revenue growth rather than by investee abatement. Target adoption carries almost no accountability, with roughly a third of expiring targets simply vanishing from disclosure without consequence.

I then derive why abatement concentrates where it does, from the condition under which a firm abates a marginal tonne: the marginal abatement cost against the carbon price actually faced, the input savings the measure delivers, and the premium a buyer will pay, less a hurdle from irreversibility. Regulation, technology and demand are the three terms in that inequality, and reading it back explains several findings that otherwise look like puzzles. It also bounds what follows, because firm characteristics can only tip decisions where a sector and region leave the economics near balance.

I then review the firm-level features that have been tested as predictors of decarbonisation, and separate them into those with stable signs across studies (regulatory exposure, committed capital, demanding management practice, target structure), those that are unstable or reverse sign (aggregate governance and ESG scores, compensation design), and those that function as warning signs rather than predictors (Scope 3 exclusion, offset reliance, misaligned lobbying, disclosure vagueness). Two structural problems cut across the whole indicator literature. Indicators saturate: as an indicator becomes universal it stops carrying information, which means predictive power decays and has to be re-estimated rather than assumed. Greenwashing indicators are close to orthogonal to each other, so the common practice of summing them into a composite score discards most of what they contain. And the statistic usually chosen to evaluate an indicator, the area under the ROC curve, is invariant to the base rate and systematically misleading here: it is highest in precisely the cells where an indicator changes fewest outcomes.

The review closes with design rules for empirical work in this area and an accompanying open codebase that implements the analyses the argument depends on.

1. Why this review

There is now a large enough body of careful empirical work on corporate climate targets that the interesting question has shifted. Five years ago the open question was whether companies would set targets at all. They did: in a sample of 1,972 large companies drawn from the Transition Pathway Initiative universe, long-term net zero commitments went from 70 in 2019 to 1,236 by 2024, or 63% of the sample (Dietz and Hastreiter 2026). Across the FTSE All-World, 70% of constituents disclosed a climate target in 2024 against 7% in 2018, and 57% disclosed a net-zero commitment against 1% (LSEG 2026).

The open question now is what any of that means for physical emissions, and by extension which observable company characteristics an investor, regulator or researcher can use to tell a firm that is going to decarbonise from one that is not. That question has a practical edge. Portfolio carbon metrics are embedded in regulatory disclosure regimes, investment mandates and, in some cases, binding decarbonisation targets. If those metrics move for reasons unconnected to physical abatement, the mandates built on them will reward the wrong behaviour.

My reading of the recent evidence is that this is largely what has happened. The review is organised around that claim. Section 2 sets out where corporate decarbonisation actually stands. Section 3 works through the measurement choices that shape nearly every published estimate in this field, because several apparent empirical findings turn out to be properties of those choices. Section 4 covers accountability. Section 5 identifies the challenges that follow. Section 6 turns to mechanism, deriving the condition under which a firm abates at all, because it bounds what any firm-level indicator can do. Sections 6 to 8 review candidate predictive indicators, and Section 10 sets out design rules that follow from the review. Section 11 lists what I think the field still needs.

A note on the evidence base. This review draws on fifteen studies, listed in the references, that I read in full. Where I make a claim I have tried to attach a number and a source to it. Where studies disagree I have said so rather than averaging them, because in several cases the disagreement is the finding.

2. Where corporate decarbonisation actually stands

2.1 Commitment has saturated, delivery has not

The adoption statistics are no longer the interesting part. Disclosure of Scope 1 and 2 emissions reached 81% of FTSE All-World constituents by count in 2024, up from 55% in 2016. Developed-market issuers are at 90%. Emerging-market disclosure rose from 41% to 73%, though that improvement is almost entirely China, where disclosure went from under 5% to over 30%; excluding China, emerging-market disclosure has barely moved since 2016 (LSEG 2026).

Set against that, the delivery numbers are modest. The median FTSE All-World constituent reduced emissions by between 0% and 1% per year outside the pandemic year, which means emissions were still rising for roughly half the index. Dispersion is wide: in 2023–24 the top quartile cut emissions by 7% or more while the bottom quartile grew them by 8% or more (LSEG 2026).

The gap between stated ambition and quantifiable commitment is visible even within the disclosure itself. Of the emissions covered by some disclosed climate target in the FTSE All-World, 9.5 Gt sit under a target of some kind, 7 Gt under a target that can be translated into an absolute reduction, and 5 Gt under a target covering 100% of in-scope emissions. Aggregating the targets that can be quantified implies Scope 1 and 2 reductions of about 25% by

2030 and 50% by 2050, which is a good deal less than a 57% net-zero adoption rate would lead one to expect (LSEG 2026).

2.2 How much of the measured decline is real

The FTSE All-World's Scope 1 and 2 emissions have been roughly flat at about 12.5 GtCO₂e since 2019, which is around 23.5% of global emissions. Over the same period every major intensity measure fell by about a third. Weighted average carbon intensity fell 5% per year between 2016 and 2024, intensity per EVIC fell 6%, intensity per market cap 5%. Global emissions per unit of GDP fell about 2% (LSEG 2026).

Decomposing those intensity declines is where the argument of this review starts. Using a logarithmic attribution that splits each change into a constituent-emissions term, a normalisation term and an allocation term, LSEG find that normalisation and allocation dominate. For the FTSE All-World over 2020–2024, constituent emissions *added* to intensity while the revenue denominator and index composition pulled it down. The corporate-bond results are starker. In the high-yield universe, aggregate emissions fell 9% per year while the chained series, which follows only issuers present in consecutive years, fell 2%. Between 2016 and 2019 allocation effects alone removed 66% of high-yield WACI as higher-intensity issuers left the index.

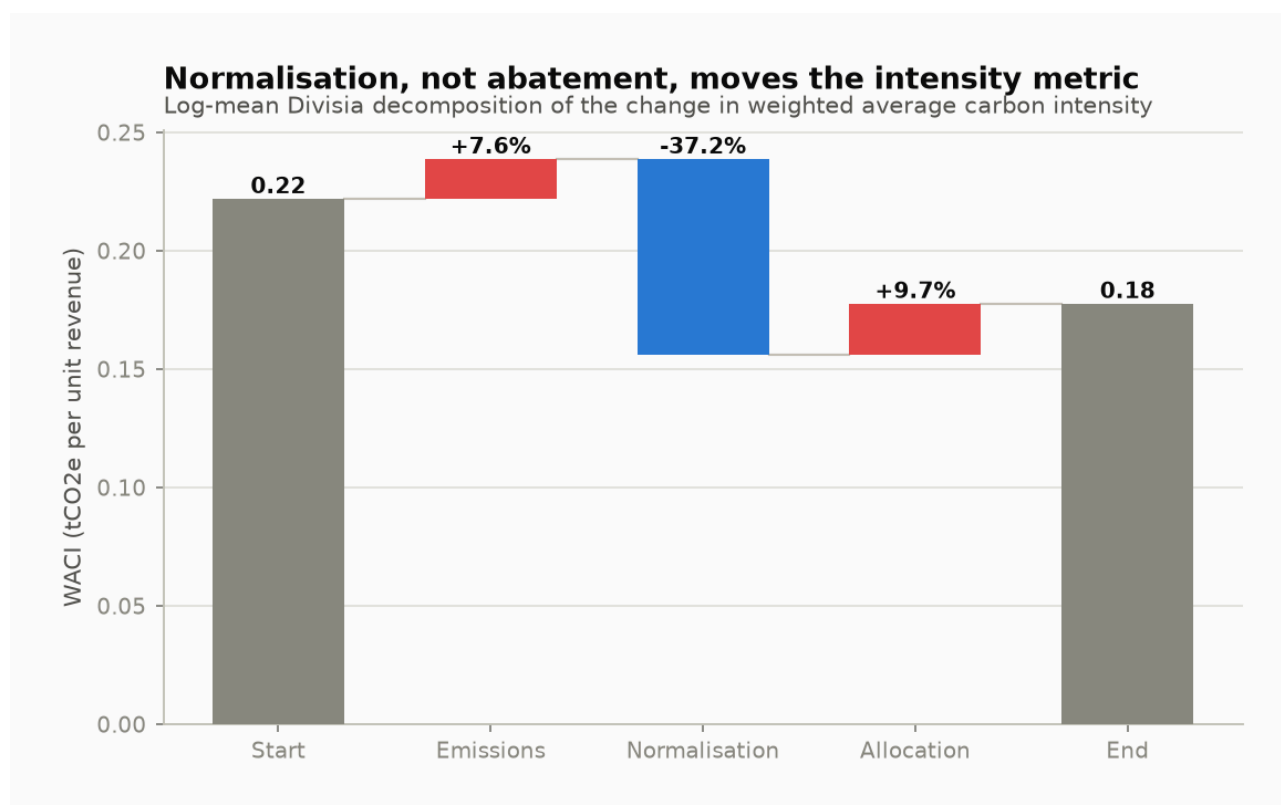


Figure 1. Log-mean Divisia decomposition of a change in weighted average carbon intensity. The normalisation term dominates while constituent emissions push the metric up. Method illustration on the synthetic panel, not an empirical estimate.

This matters beyond portfolio accounting because it establishes the general point: a carbon metric can fall a long way without any company emitting less. The chained series is the useful construct here, and it transfers directly to firm-level work. Holding the perimeter constant separates what companies did from what the sample did.

2.3 The renewable electricity wedge

The second component of the argument comes from two studies that arrive at the same place by different routes.

Ruiz Manuel and Blok (2023) evaluated 102 of the largest members of the Science Based Targets initiative and RE100 by revenue, using disclosed environmental data for 2015–2019. Collective Scope 1 and 2 emissions fell 35.6% from a baseline of 808.7 MtCO₂e, which looks like strong evidence that these initiatives work. The disaggregation is less encouraging. Most of that reduction is concentrated in eight emission-intensive companies. Most members show little evidence of emissions reductions inside their own operations and achieve progress through renewable electricity purchases. Of the renewable energy purchased, 71% came through sourcing models with low additionality, principally unbundled energy attribute certificates and utility green premiums, which are unlikely to cause additional renewable generation.

Schüder and Zülch (2026) reach a compatible conclusion from an entirely different design. They apply coarsened exact matching to 3,113 listed firms in Europe, North America and Asia over 2015–2024, and track carbon performance for up to four years after SBTi participation. SBTi participation is associated with significant reductions in Scope 1 carbon intensity in every post-participation year, with coefficients from -0.368 at $t+1$ to -0.511 at $t+4$. For Scope 2 the result splits by accounting method. Market-based Scope 2 shows statistically significant reductions at $t+2$ and $t+3$. Location-based Scope 2 shows coefficients that are consistently negative but fall above the 10% threshold in most specifications. Scope 3 shows no measurable effect at all.

Location-based Scope 2 reflects the emissions intensity of the grid a company actually draws from. Market-based Scope 2 reflects the contracts and certificates it has bought. When an effect is present in the second and absent in the first, the most economical explanation is procurement rather than abatement. That Ruiz Manuel and Blok independently document low additionality in exactly those procurement channels makes the reading harder to avoid.

The LSEG data show the same wedge opening in the sector where electricity demand is growing fastest. Among FTSE All-World constituents that consistently disclosed both measures from 2020 to 2024, Technology's location-based Scope 2 rose 60% while its market-based Scope 2 rose 22%. Technology Scope 1 and 2 emissions rose 28% between 2019 and 2024, more than any other sector, and Scope 2 is about 84% of that sector's total.

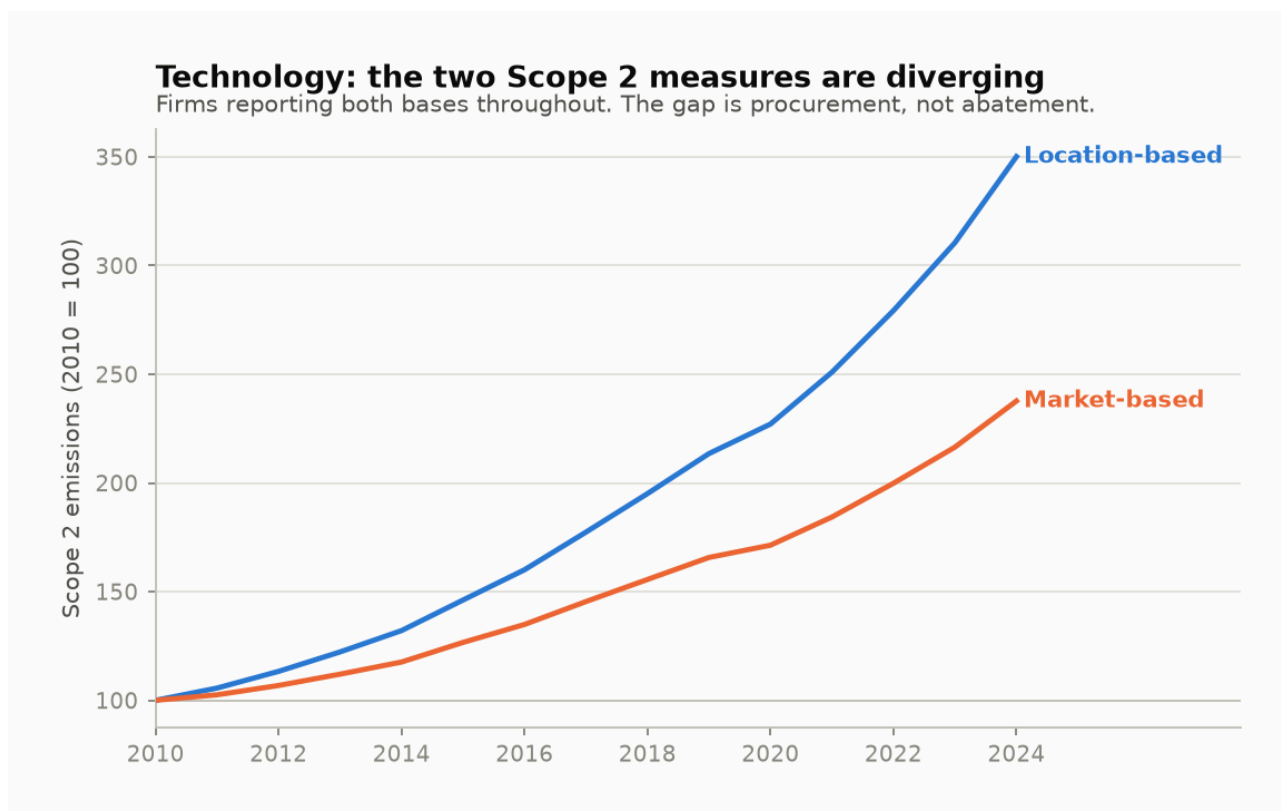


Figure 2. The two Scope 2 measures for firms reporting both throughout. Market-based reporting absorbs procurement; location-based tracks the grid. Method illustration on the synthetic panel.

Three independent sources, then, point at the same seam in the accounting system. I take this to be the central empirical fact about corporate decarbonisation as currently measured.

3. Five measurement choices that shape the findings

Before turning to predictors it is worth being explicit about the measurement decisions that sit underneath every estimate in this literature. Several published findings are better understood as consequences of these decisions than as facts about firms.

3.1 Scope 2 accounting method

Covered above. The practical implication for research design is that a study which pools location-based and market-based Scope 2, or which uses whichever the company reported, is partly measuring procurement policy. Any firm-level decarbonisation label needs the method pinned, and location-based is the conservative choice if the question is physical abatement. It is worth noting that the divergence will widen: electricity demand growth from data centres is running ahead of clean procurement in some markets, and the two series have no reason to reconverge.

3.2 Scope 3

Scope 3 reporting has improved in volume and not in usefulness. Across the FTSE All-World, 61% of companies disclosed at least one Scope 3 category in 2024, up from 35% in 2016, but only 40% disclosed a material category. The composition of disclosure is close to inverse to materiality. Category 6, business travel, is the most

frequently disclosed at 44% of firms and accounts for under 1% of reported Scope 3 volume. Category 11, use of sold products, accounts for 21.4% of volume and is disclosed far less often (LSEG 2026).

Schüder and Zülch find no SBTi effect on Scope 3. Dietz and Hastreiter test whether firms whose net zero targets include Scope 3 reduce emissions more than those whose targets do not, and find no statistically significant difference, though the coefficients are negative. Brown, Hsu and Manya find that lack of Scope 3 coverage is the single most common greenwashing red flag, present in 70% of pledging companies.

The reasonable conclusion is that Scope 3 currently cannot support cross-sectional classification of firms. It can support within-sector comparison in sectors where the material category is well defined and widely reported, and it should be treated as missing elsewhere rather than imputed.

3.3 Reported against estimated data

This distinction does more work than it usually gets credit for. LSEG's attribution splits the emissions contribution by data source precisely because estimated and reported emissions behave differently. Their own caveat on the Transition Pathway Initiative result is instructive: the dispersion in the lowest management-quality band is the widest of any band and is over-represented at both extremes, but much of that reflects smaller companies with estimated emissions and largely disappears among disclosed emitters.

Bingler et al. (2024) hit the same issue from the other side and are unusually candid about it. Their cheap talk index correlates with emissions growth in third-party estimated data from Urgentem but shows no correlation in self-reported data. They interpret this as firms with high emissions growth being both less likely to disclose and more prone to vague language, and they deliberately avoid basing the analysis on self-reported data because cheap talk and self-report reliability are likely to be positively correlated, which would confound the estimate. That is the right call. It also means the headline result rests on a vendor's estimation model, and if that model uses disclosure characteristics as inputs then part of the correlation is mechanical. The authors do not claim otherwise, but users of the result should hold it more loosely than the abstract suggests.

3.4 Denominator choice

Intensity metrics divide by something, and the something moves. Revenue denominators move with price levels, commodity cycles and inflation. EVIC and market capitalisation move with valuations and, in EVIC's case, reward leverage. Physical denominators are the most defensible measure of carbon efficiency and exist for only a minority of sectors (LSEG 2026). No choice is neutral, and the normalisation term in LSEG's attribution is frequently larger than the emissions term.

3.5 Metric divergence

Fliegel (2026) assembles eight firm-level transition risk metrics, covering reported EU taxonomy alignment of capex and revenues, emission intensities, Refinitiv and MSCI E-scores, the Refinitiv Business Classification, and text-based measures, and computes rank correlations. Within-group correlations are positive but not high: 0.73 for taxonomy-based measures, 0.60 for emission-intensity-based, 0.58 for text-based. Between-group correlations are close to zero. Most striking, all taxonomy-based proxies correlate *negatively* with inverted emission intensity metrics, meaning that firms scoring well on one family of measures tend to score badly on the other.

Fliegel then evaluates the metrics by the return sensitivity of portfolios built on each to market-wide transition risk news. Taxonomy alignment and sector/technology classification perform best for green portfolios; MSCI E-scores and Refinitiv Business Classification perform best for brown firms. His conclusion is that forward-looking

measures such as taxonomy capex alignment outperform pure exposure measures, and that findings resting on a single transition risk metric should be treated with caution.

For anyone building a predictive model, the practical consequence is that "transition risk" is not one variable. Choosing a different proxy will produce a different answer, and robustness across proxies is the minimum standard.

4. Targets are not policed

Jiang, Kim and Lu (2025) examined the 1,041 firms whose emissions targets expired in 2020. Of those, 88 (9%) failed and 320 (31%) disappeared, meaning the target vanished from disclosure without a stated outcome. Only three of the failed firms received any media coverage. After a firm misses its target the authors find no significant market reaction, no change in media sentiment, no change in environmental scores, and no change in environment-related shareholder proposals. Announcing the target, by contrast, is rewarded with significant improvements in media sentiment and environmental scores.

The asymmetry is the finding. There is a reward for announcing and no penalty for missing, and the most common outcome is neither success nor failure but silence. Any model that treats target adoption as a costly signal is mispricing it, because in this sample it was not costly.

Bolton and Kacperczyk (2025) reach a complementary conclusion on selection. Firms that commit do subsequently reduce emissions, but the firms that commit, and those that make the most ambitious commitments, tend to be the ones with lower emissions already. Firms are less likely to make an ambitious commitment when their Scope 1 emissions are higher. Their summary is that the commitment movements "have been successful in drawing the willing but have found greater resistance from the companies that most need to reduce their emissions". The effect on overall emissions, including non-committing firms, has been small.

Dietz and Hastreiter (2026) test the near-term behavioural consequences of long-term net zero adoption directly, using staggered difference-in-differences with matching on the TPI universe. They find little evidence of large or immediate emissions reductions. Emissions coefficients are generally negative and consistent with gradual reduction, but noisy and imprecisely estimated.

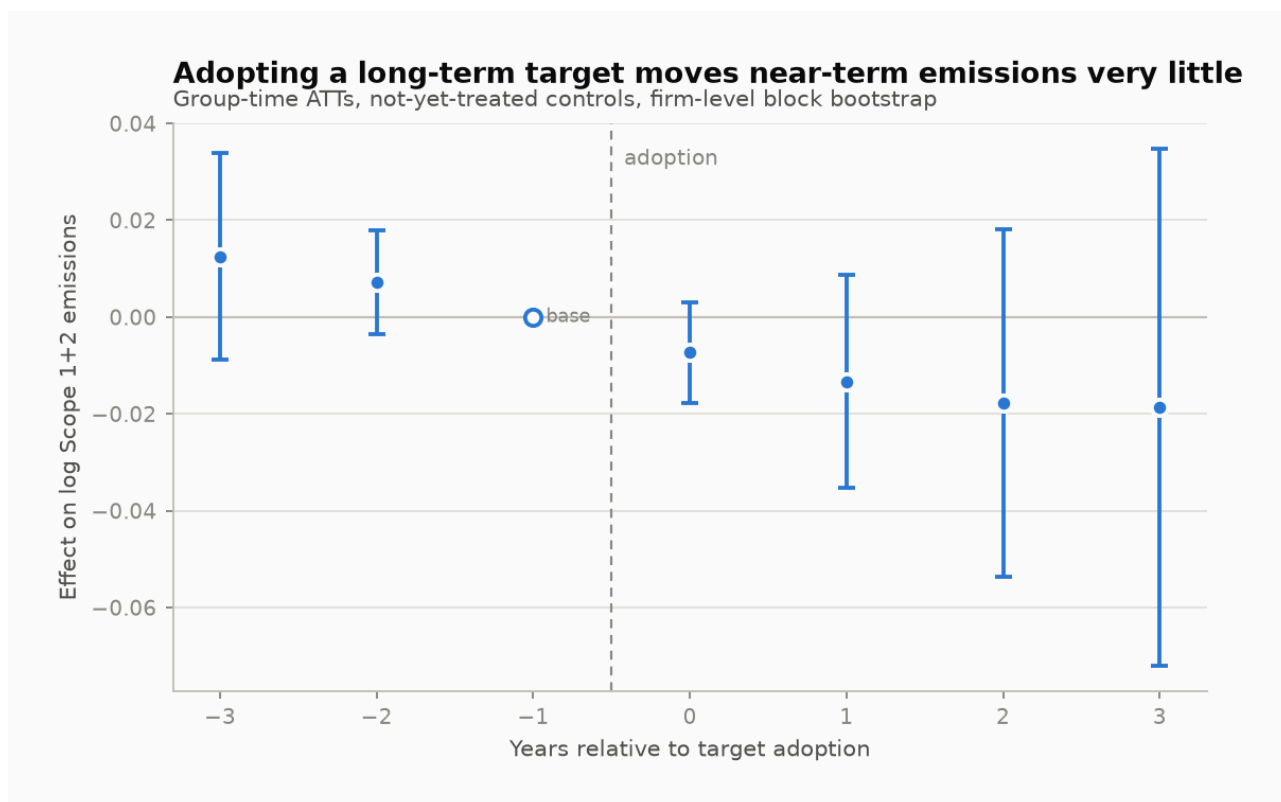


Figure 3. Group-time average treatment effects of long-term target adoption on log Scope 1+2 emissions, with not-yet-treated controls and a firm-level block bootstrap. Estimates are negative and indistinguishable from zero. Generated on a synthetic panel in which adoption has no causal effect and adopters are selected on an unobserved propensity, so this is a demonstration that the estimator recovers the null where a naive comparison would not.

Their governance results are the more interesting part, and I return to them in Section 7.3 because they bear on indicator design rather than on accountability.

5. What follows for the next five years

Four problems seem to me to follow from the evidence above.

Electricity demand growth is about to test the Scope 2 convention. Technology sector emissions are rising faster than any other sector's, driven by data centre load, and the gap between location-based and market-based reporting is already 38 percentage points over 2020–2024 for dual reporters. If clean procurement keeps pace, market-based emissions stay contained while physical grid emissions rise. If it does not, today's demand growth becomes tomorrow's reported growth. Either way, investors relying on market-based figures will see the problem late.

The quantifiable target base is much smaller than the headline. Roughly half the emissions nominally covered by targets sit under targets that either cannot be converted into an absolute reduction or do not cover full scope. Regulatory frameworks that count target adoption rather than target structure will keep producing coverage statistics that overstate committed abatement.

Indicator saturation is eroding the discriminating power of the standard measures. This is developed in Section 10, but the short version is that when 70% of an index has a target and 81% disclose emissions, those

variables no longer separate firms. Dietz and Hastreiter provide direct evidence of the mechanism.

Scope 3 remains unusable for cross-sectional work and is where the majority of value chain emissions sit for most non-utility sectors. Nothing in the current disclosure trajectory fixes this before 2030.

6. Why abatement happens where it does

The sections above describe what has happened. This one asks why, because the answer bounds what any firm-level indicator can possibly do, and the sections that follow cannot be read correctly without it.

6.1 One inequality

A firm abates a marginal tonne when the value it captures covers what abatement costs it:

$$\text{abate iff } \text{MAC} \leq p \cdot \kappa + s + \pi - h$$

where MAC is the marginal abatement cost, $p \cdot \kappa$ the carbon price times the share of the firm's marginal emissions actually priced, s the input cost savings the measure itself delivers, π the revenue premium per tonne from customers paying more for the low-carbon product, and h a hurdle arising from irreversibility under uncertainty. Call the difference the **abatement gap**:

$$G = \text{MAC} - (p \cdot \kappa + s + \pi) + h$$

Firms abate where $G < 0$. Regulation, technology and demand are simply the three terms that push it down. Nothing in this literature escapes the inequality: targets, governance and disclosure operate by shifting a firm's position relative to it, never by suspending it.

This reframes the concentration result from Section 2. Ruiz Manuel and Blok find 86% of member abatement in eight electricity and heavy-industry firms, with 98.6% of all internal Scope 1 reductions. The natural reading is that those firms were better managed. The more economical one is that G was already negative in their cell and positive nearly everywhere else.

6.2 Regulation: the effective marginal price

Two features of $p \cdot \kappa$ matter more than the headline carbon price.

Free allocation design decides whether the price bites, and how. Lump-sum or grandfathered allocation is an infra-marginal transfer: it changes the firm's wealth, not its marginal incentive, so abatement incentives are preserved intact. Output-based or benchmarked allocation ties allowances to production, which functions as an output subsidy alongside the carbon price. The firm retains a full incentive to cut emissions *per unit of output* and a weakened one to cut emissions *absolutely*.

That has a signature this review has already documented: intensity falling while absolute emissions stay flat. Composition and revenue growth explain most of that gap, as Section 2.2 sets out, but output-based allocation is a second mechanism pushing the same way, and unlike composition it is deliberate policy design rather than an accounting artefact.

Colmer et al.'s null Phase I result needs no behavioural explanation. Phase I over-allocated, the price collapsed toward zero, so $p \cdot \kappa \approx 0$ and G never turned negative. The policy did not fail to change behaviour; it failed to create a price. The measurement implication is direct: a binary "covered by an ETS" indicator encodes $\kappa > 0$, which is not the variable in the inequality. A firm inside a scheme with a collapsed price is, for predictive purposes, unregulated.

Credibility enters through the variance, not the mean. Abatement capital is irreversible and long-lived, so what matters is the expected carbon price across decades. Under uncertainty an irreversible investment carries an option value of waiting, and the firm rationally requires expected value to exceed cost by a margin rather than merely equal it. That margin is h . It follows that a legislated price floor or a credible long-dated trajectory can trigger investment by narrowing the distribution of expected prices without raising its mean at all, that a credible future constraint moves investment years before it binds, and that price volatility is a genuine cost of market-based instruments which headline-rate comparisons miss entirely.

6.3 Technology: learning is a property of manufacturing multiplicity

Technology costs follow cumulative production rather than calendar time, falling a roughly constant fraction per doubling: about 24% for solar PV and 20% for battery packs, giving declines of 89% and 86% respectively since 2010. This is a positive feedback loop, since deployment lowers cost which raises deployment, and systems with positive feedback do not respond proportionally to a forcing. They sit still and then tip. That is why sectoral abatement looks binary rather than graduated, and why extrapolating a sector's past rate is a poor forecast on either side of its tipping point.

The reason cement does not get a learning curve is that the learning denominator is *units manufactured*. Solar modules have been produced billions of times; a cement kiln with carbon capture a handful. The available doublings differ by orders of magnitude. Steep learning is therefore not a property of clean technology but of modular, mass-manufactured goods, and its absence is a property of bespoke capital projects. Over the same period in which clean modular goods fell 64 to 89%, thermal plant construction costs *rose* by roughly 28%.

The useful consequence is that **electrification transfers learning between sectors**. A sector that can electrify inherits a 20 to 24% learning rate it did nothing to earn; it buys the product of someone else's cost declines. A sector that cannot, whether for energy density as in aviation, process temperature, or process chemistry as in cement calcination, must fund its own learning with far fewer doublings available. The decisive technology question for a sector is therefore not whether a green option exists but whether the process can be electrified.

This also explains an apparent paradox in Section 2. Technology-sector emissions are rising faster than any other sector's while its firms are among the most engaged on climate. Their emissions are overwhelmingly Scope 2, their abatement is the grid's problem rather than theirs, and their demand growth is outrunning the grid's improvement.

6.4 Demand: a missing market rather than a missing preference

The premium term is where analysis usually stops at the observation that customers will not pay. The mechanism is more specific.

Abatement cost is concentrated upstream in materials and energy; willingness to pay sits downstream with the final consumer. A €150 per tonne green steel premium is roughly €100 to €150 on a car containing about a tonne of steel, which is immaterial against a €30,000 vehicle and decisive against a steelmaker's margin. **The premium**

is negligible where the money is and decisive where the cost is, so position in the value chain rather than intensity of preference determines who can absorb it. That is why automotive offtake agreements at \$30 to \$80 a tonne exist at all.

Three failures stop a premium forming without institutional help. Low-carbon steel is physically identical to ordinary steel, making it a **credence good** whose attribute the buyer cannot verify, so certification is a precondition for the market rather than an accessory to it. The first buyer to pay a premium funds the deployment that moves the producer down its cost curve, and competitors then buy cheaper, so **first movers are undersupplied relative to the social return**. And no liquid market exists for differentiated low-carbon commodities, so price discovery happens bilaterally, which is why the evidence on premiums arrives as contract disclosures rather than published prices.

It follows that the premium is the most regionally variable of the three terms and the one most directly manufactured by policy. The clearest illustration is Chinese steel: a production cost gap around \$140 a tonne against a willingness to pay capped near \$20, roughly a seventh of the gap, for the same technology and the same physics as a European producer facing €120 to €180 premiums. The EU's border adjustment mechanism is best understood in exactly these terms, as converting the premium from something a buyer may choose to pay into something an importer must, which changes a sector's economics far more than its technology does.

Note finally that electricity required no demand channel at all. Electrons are fungible, so a clean generator sells into the same market at the same price. That is a third reason the sector moved first, alongside early pricing and steep learning: it is the one sector where decarbonising required no customer to pay more for a differentiated product.

6.5 Why the three behave as complements

At the margin the three terms are substitutes, entering the inequality additively, and any one of them closing the gap suffices. Empirically they behave as complements, and the reason is **h** combined with lumpiness. A firm does not buy 3% of an electric arc furnace. It faces a discrete, irreversible, long-lived commitment that must clear the cost with a margin for the option value of waiting. For hard-to-abate sectors no single channel is large enough to clear a MAC of \$150 to \$300 a tonne on its own: a \$60 carbon price does not do it, a 10 to 25% premium on a thin slice of demand does not do it, and the technology has no learning curve to ride. All three must move together, so they appear multiplicative in the data though the underlying model is additive.

There is also a dynamic complementarity with no static counterpart. Deployment funded through any channel today lowers MAC tomorrow through learning, which lowers the carbon price and premium needed thereafter. Policy that subsidises early deployment is buying down the future cost of every other instrument.

6.6 What this bounds

The inequality places a sharp limit on the sections that follow. Where **G** is large and negative every firm abates, and where it is large and positive none does; in both cases there is no variation left for a firm-level indicator to explain. Firm characteristics such as capital access, management quality and planning horizon tip the decision only where the cell's economics leave it near balance.

Simulating a firm-quality indicator across the gap distribution makes the shape explicit, and exposes a trap in the statistic:

cell gap \$/tCO ₂ e	share abating	AUC	risk difference
-120	100%	0.500	0.000
-60	98%	0.646	0.025
-30	85%	0.630	0.151
0	50%	0.634	0.301
+30	15%	0.660	0.184
+60	2%	0.715	0.038
+120	0%	0.500	0.000

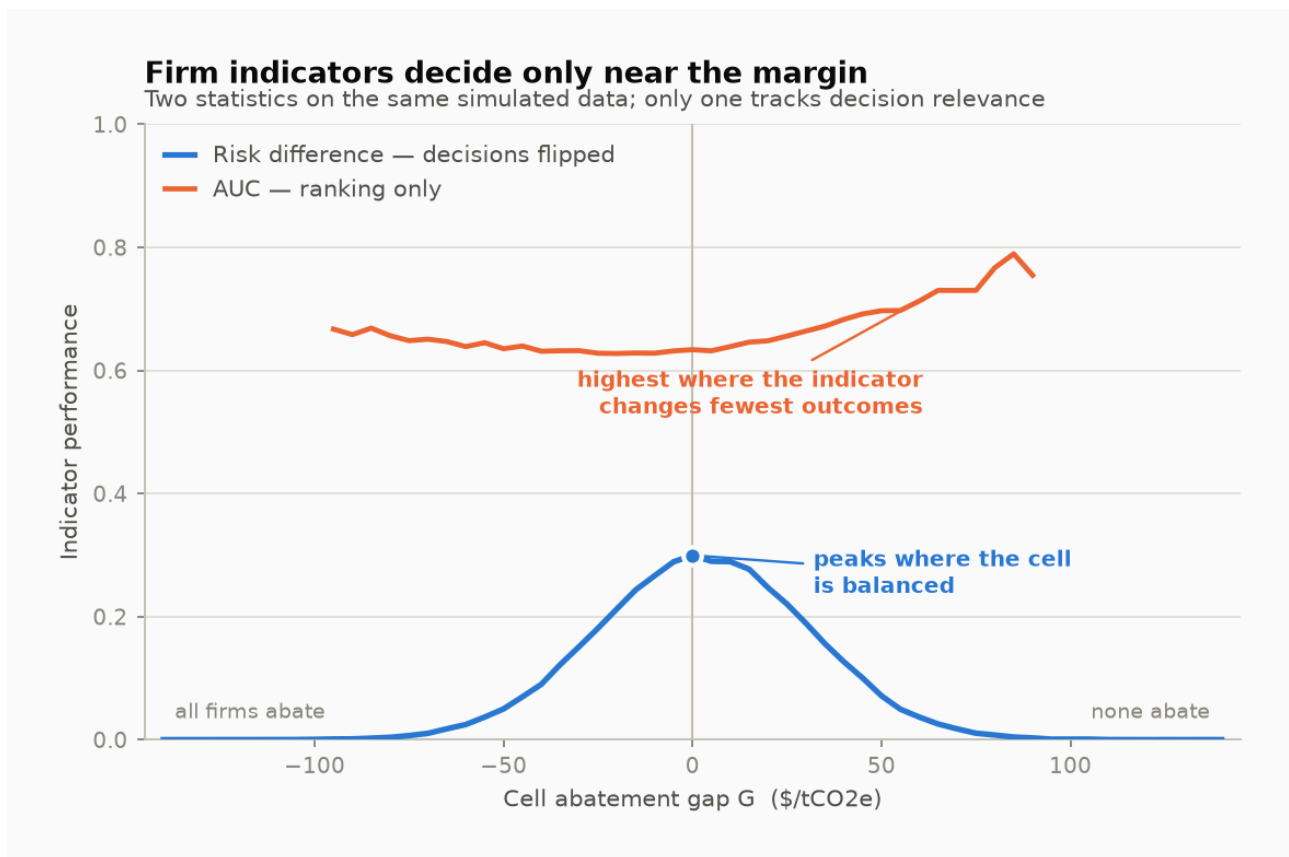


Figure 7. Two statistics on the same simulated data, across the abatement gap. The risk difference traces decision relevance; the AUC does not, and is highest where the indicator changes fewest outcomes. AUC is shown as undefined where one class is empty. Method illustration on simulated data.

Decision relevance is an inverted U, peaking where the cell is balanced and collapsing at both extremes. The risk difference traces it; **the AUC does not**, instead rising to 0.715 at a gap of +60 where the indicator changes almost no outcomes. AUC is rank-based and invariant to the base rate, so where few firms abate the handful that do are strongly selected on quality and the ranking looks excellent while nothing is decided. A study reporting AUC alone would conclude an indicator works best in hard-to-abate sectors, which is the opposite of where it matters.

This is also a candidate explanation for why the literature disagrees with itself about firm-level predictors. Studies sampling different sectors and periods sample different parts of the gap distribution, and correct analyses of the same underlying process will reach different conclusions.

7. Indicators with a stable relationship to decarbonisation

I group candidate predictors by how much confidence the current evidence supports. The criterion is not effect size but whether the sign is stable across studies with different samples and designs.

7.1 Carbon pricing exposure

This is the best-identified result in the field. Colmer, Martin, Muûls and Wagner (2025) use administrative data on French manufacturing firms to estimate that the EU Emissions Trading System caused regulated firms to reduce CO₂ emissions by 14–16%, with no detectable contraction in economic activity, no evidence of outsourcing to unregulated firms or markets, and targeted investment reducing the emissions intensity of production.

Two features of the result deserve emphasis. First, the effect is concentrated in the phases where the scheme actually bit; point estimates for Phase I are close to zero and statistically insignificant. Regulatory exposure only predicts abatement when the regulation is stringent, which means a binary "covered by an ETS" feature is too coarse. Second, the authors rationalise the absence of negative economic effects through inattention: firms with low initial productivity or high energy intensity underinvest in energy-saving capital before regulation, and show larger emissions reductions and increases in economic activity afterwards. If that mechanism generalises, the firms where regulation produces the most abatement are identifiable in advance.

7.2 Committed capital

Fliegel's evaluation puts EU taxonomy capex alignment among the strongest performing transition metrics, and his general conclusion favours forward-looking measures over exposure measures. The logic is straightforward: capital expenditure is a decision that has already been made and paid for, which makes it harder to reverse than a stated target.

The limitation is coverage. Taxonomy capex alignment is reported under a regime that applies to a subset of firms in one jurisdiction, so any global model faces systematic missingness correlated with region. Missingness of this kind is informative and should be modelled as such rather than imputed.

Frisch et al. (2025) make a compatible argument from a qualitative direction. Their framework treats the core business through three dimensions, management, value chain and investments, with three categories each, and their central claim is that decarbonisation should be assessed by how deeply it is integrated into the core business rather than by counting climate management activities in isolation. Investment allocation is the dimension in their scheme that is hardest to fake.

7.3 Demanding management practice, weighted by rarity

This is the most useful methodological finding in the corpus and it comes from Dietz and Hastreiter (2026).

They measure climate management using TPI Management Quality indicators, and construct three versions: a count of satisfied indicators, a hierarchical level, and a weighted score that gives greater weight to less commonly implemented and more demanding practices. The three behave differently after long-term net zero adoption. The count shows no systematic effect. The level shows a *negative* post-adoption effect, statistically significant at t+3. The weighted score shows positive, increasing and statistically significant effects.

The explanation they give for the negative result on levels is that non-adopters caught up with adopters on basic management practices. Once nearly everyone has a board committee and a disclosure process, having one stops

distinguishing you, and a measure built on counting them will show adopters converging toward the mean.

Decomposed by TCFD theme, the pattern holds: the Governance theme, which contains the less ambitious indicators, shows negative post-adoption effects, while the Strategy theme, which contains the more demanding ones, shows positive and increasing effects. The Strategy result is preceded by a significant coefficient in the year before adoption, which the authors attribute to anticipation, and cohort analysis locates that pre-trend in the 2021 cohort specifically.

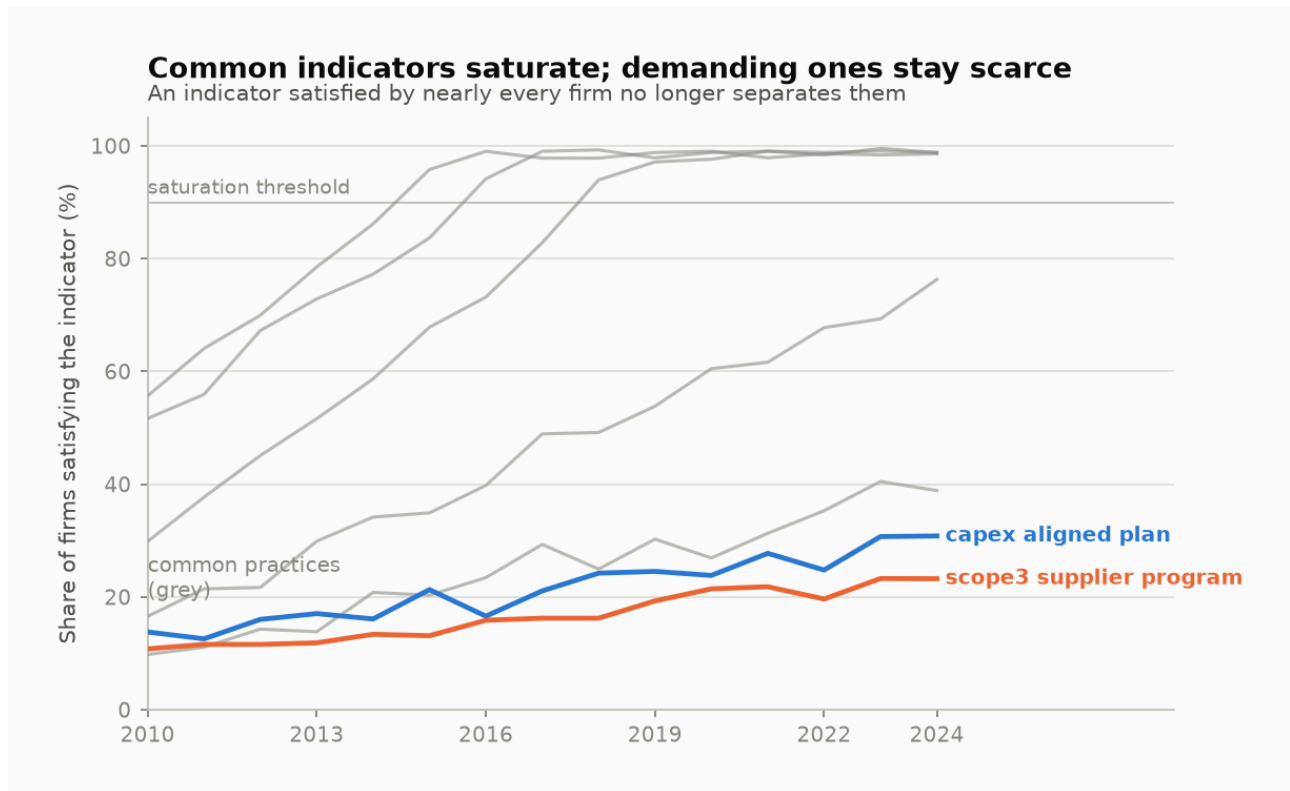


Figure 4. Indicator prevalence over time. Once a practice is satisfied by nearly every firm it stops separating them, which is why a raw count of indicators can lose discriminatory power while a rarity-weighted score retains it. Method illustration on the synthetic panel.

Two lessons. Rarity-weighting a governance indicator set recovers signal that a raw count destroys. And the timing of improvement runs slightly ahead of the announcement, which means event studies keyed to announcement dates will understate the association.

7.4 Target structure

Target design carries information that target existence does not. The LSEG funnel from 9.5 Gt to 7 Gt to 5 Gt is one form of evidence. Bolton and Kacperczyk note that absolute reduction commitments differ from intensity commitments in a way that matters, since intensity targets permit total emissions to rise. Dietz and Hastreiter find weak, directionally negative evidence that targets including Scope 3 are associated with larger subsequent reductions, though not statistically significant.

Silvia et al. (2026) provide the most direct test. Using 239 Fortune Global 500 non-financial firms and 1,126 firm-year observations over 2018–2023, they construct a Climate Transition Plan credibility index and find that higher credibility is significantly associated with lower subsequent carbon emission changes. External assurance strengthens the association; mandatory or quasi-mandatory disclosure environments provide more modest

moderating support. The Scope 3 subsample produces weaker evidence, in line with everything else in this review. The design is observational panel, and the authors are appropriately careful about causal language.

7.5 Past emissions

Any model in this area has to beat a persistence baseline. Emissions trajectories are highly autocorrelated, and the strongest single predictor of next year's emissions is this year's. None of the studies in this corpus is primarily a forecasting paper, but the point surfaces repeatedly: Bolton and Kacperczyk control for prior emissions changes over one, three and five years; Schüder and Zülch match on pre-treatment characteristics; Dietz and Hastreiter match on emissions trajectory.

The relevant caution concerns Xu, Wei and Ji (2026), which is the one explicitly predictive study in the corpus. They compare six models on listed firms over 2016–2023 and find XGBoost best, with R^2 of 0.95 in the pre-COVID period using all variables and 0.85 excluding carbon-related variables. Within the governance category, board characteristics carry the highest importance, and with the carbon signal removed the model shifts weight onto liquidity, profitability and firm value.

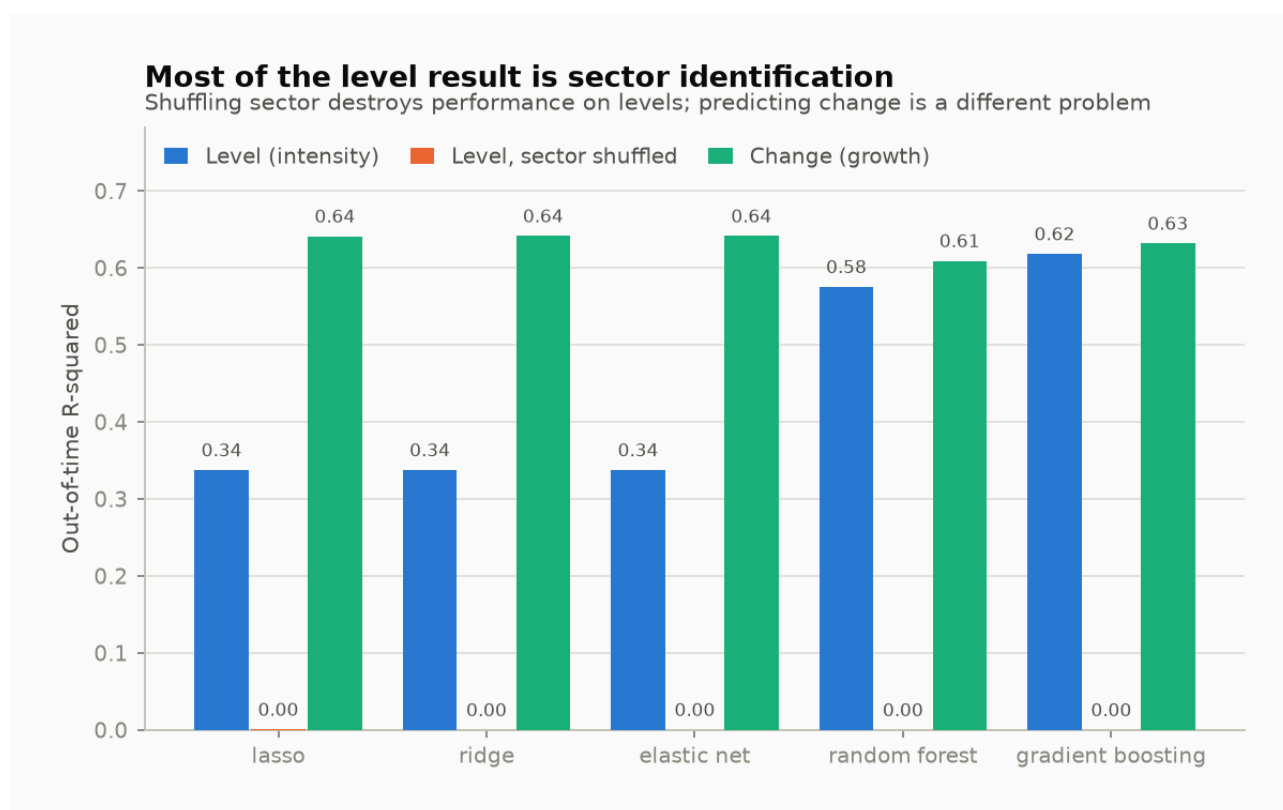


Figure 5. Out-of-time R-squared for the same feature set against a level target and a change target, with a sector-shuffled control. Shuffling sector destroys performance on levels, which is what most of that performance was. Method illustration on the synthetic panel.

The result is interesting but it should not be read as evidence that governance predicts decarbonisation. Their target is carbon emission *intensity*, a level, not a change. An R^2 of 0.85 predicting intensity levels without carbon inputs is mostly the model recovering sector and size, which are the dominant determinants of how carbon-intensive a firm is. Predicting which firms are dirty is a different and much easier problem than predicting which firms are getting cleaner. The distinction is easy to lose and it separates most of the machine learning literature in this area from the question investors actually have.

8. Indicators that flag the absence of transition

A separate literature has developed around detecting the opposite condition. These measures are not predictors with the sign reversed; they behave differently and should be handled differently.

8.1 The red flag framework

Brown, Hsu and Manya (2026) built the largest empirical assessment of its kind, combining CDP, InfluenceMap and Net Zero Tracker data across 4,131 companies, of which 3,574 made some climate claim. They assess seven dimensions of greenwashing risk. Of the companies making a pledge, 96% exhibit at least one risk indicator.

Prevalence by dimension:

Red flag	Share of pledging companies
Lack of Scope 3 coverage	70%
Questionable use of carbon offsets	40%
No interim targets	21%
Lack of progress toward targets	20%
No implementation plan	18%
Neutrality claim on a CO ₂ -only or unspecified gas inventory	11%
Lobbying inconsistent with commitment	10%

One dimension is routinely misread, including in the paper's own results text, which describes it as a failure "to comprehensively address all emission scopes". Their Methods section is explicit that it concerns *gases*: the flag fires when a company claims net zero, GHG neutrality or similar while its inventory covers only carbon dioxide or does not say which gases it covers. A company with a plain percentage-reduction target is not caught by it at all, because there is no neutrality claim for the incomplete inventory to contradict.

Two-fifths of companies show exactly one flag, and 12% are flagged on four or more. Restricting to firms with explicit net-zero targets barely moves the overall incidence (95.8%) but changes the composition substantially: questionable offset use is 29 percentage points higher, absence of an implementation plan 10 points higher, while lack of Scope 3 coverage is 22 points lower.

Regional variation is modest overall, at 95% in Europe and the Global South against about 97% in North America and East Asia and the Pacific, but the lobbying indicator is noticeably less prevalent among European firms.

8.2 The orthogonality problem

The most consequential result in that paper is easy to miss, and the authors act on it themselves: they decline to aggregate the seven dimensions into a single index at all. Their stated reasons are that no agreed weighting approach exists, that any weighting would be "inherently subjective", and that binary indicators with no intensity dimension "could dilute their meaning" once combined. The supporting evidence is that the seven indicators are only weakly correlated with each other, and several of the pairwise phi correlations are negative. Higher target

ambition is *negatively* associated with Scope 3 gaps ($r = -0.19$) and with offset reliance ($r = -0.18$). Being off-track shows near-zero correlations with almost everything.

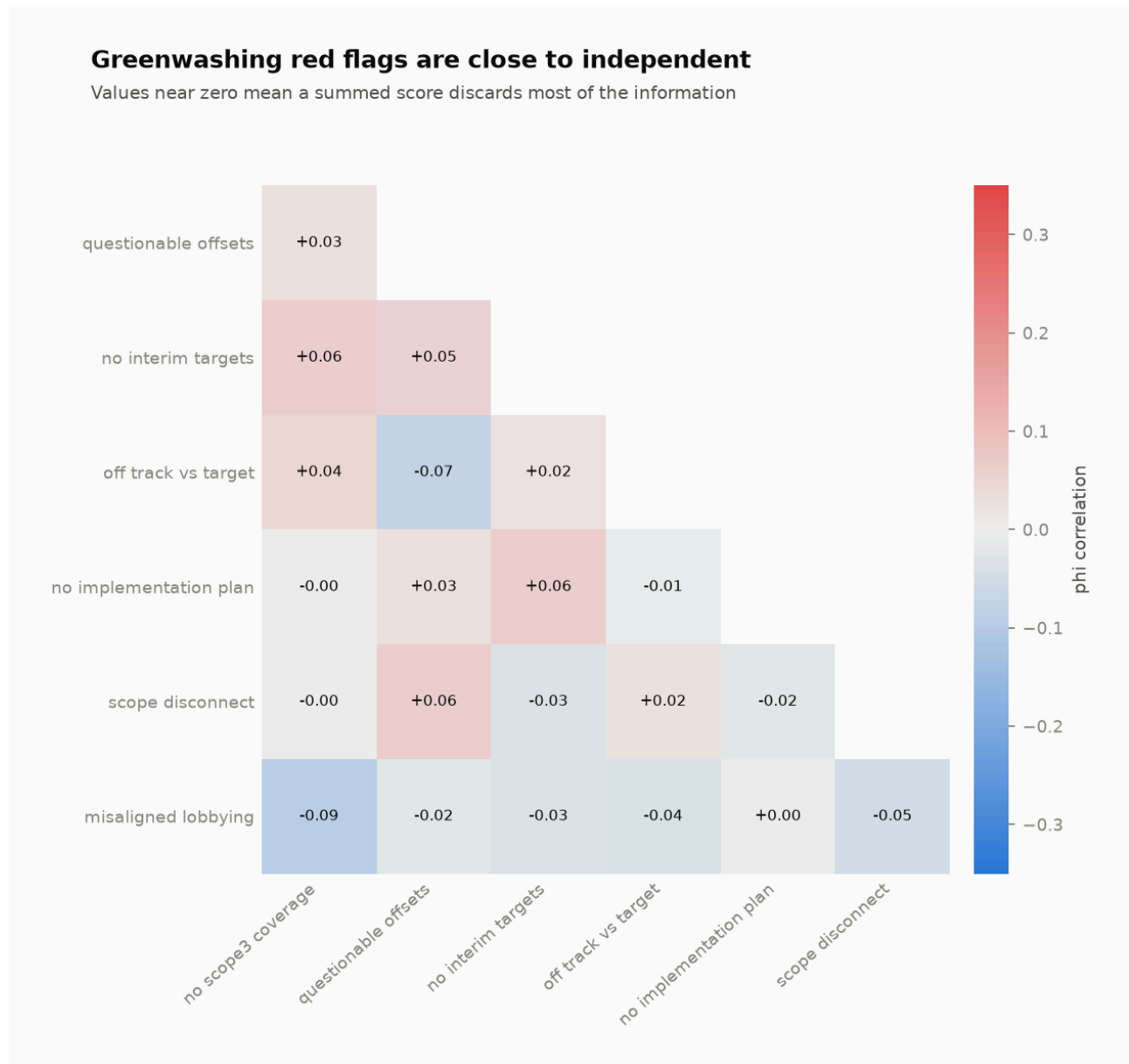


Figure 6. Pairwise phi correlations between the seven red flags. Values cluster near zero and several are negative. Method illustration on a synthetic panel calibrated to the prevalences Brown, Hsu and Manya report.

Greenwashing, on this evidence, is not a single latent trait that a composite score can measure. A firm with an ambitious target and heavy offset reliance and a firm with a narrow target and no offsets are different objects, and summing flags makes them look identical. The framework is better used as a profile, and the analytically interesting question is which combinations occur together and what each combination predicts.

8.3 Disclosure language

Bingler et al. (2024) fine-tune ClimateBERT to build ClimateBertCTI, a classifier for climate-related cheap talk, and construct a firm-level cheap talk index from annual reports of MSCI World constituents over 2010–2020. Four results: targeted climate engagement is associated with less cheap talk; supporting voluntary disclosure frameworks is associated with *more* cheap talk; cheap talk is associated with higher emissions growth; and cheap

talk is associated with more negative news coverage and controversies. The utility sector's index rose by almost 100% over the period.

The caveats from Section 3.3 apply, and there is a second one: the emissions growth relationship is insignificant over the full sample period and emerges in the later years, which the authors attribute to increased public awareness after 2015. A relationship that appears only in a subperiod, only in one data source, deserves replication before it goes into a production model.

8.4 Target detection at scale

Schimanski et al. (2023) provide the tooling layer. ClimateBERT-NetZero is trained on an expert-annotated set of 3.5K text samples to classify whether a passage contains a net zero or reduction target, reaching over 96% accuracy and outperforming larger general-purpose models on the task. Combined with question-answering models it can extract the ambition expressed in a detected target, and the authors demonstrate it on quarterly earnings call transcripts.

This matters for indicator construction because the target-structure variables in Section 6.4 are expensive to collect by hand. A classifier that separates net zero claims from quantified reduction targets at scale is the input that makes structural target features tractable across a large universe.

9. Indicators that do not behave as expected

It is worth being blunt about the measures that do not survive contact with the evidence, because several are widely used.

Aggregate governance and ESG scores. Oyewo (2023) analyses 336 multinationals across 42 non-financial industries and 32 countries over 15 years and finds relationships that are both curvilinear and, in places, opposite in sign to the standard expectation. Board gender diversity, CEO duality and the presence of an ESG committee are negatively associated with carbon emissions rate, which is the expected direction. Board independence and ESG-based compensation have significant *positive* associations, meaning firms with more independent boards and with ESG-linked pay had worse carbon performance.

Schüder and Zülch find the same sign independently. In their Scope 1 models the governance control carries a positive and significant coefficient (0.007 to 0.008, $p < 0.01$ in three of four specifications), again meaning higher governance scores went with higher carbon intensity.

I do not think either paper establishes that good governance causes emissions. The more likely reading is that governance scores are correlated with size, sector and disclosure propensity in ways that survive the controls used. But that is precisely the problem: a variable whose sign flips depending on what else is in the model is not a usable predictor, and the widespread practice of including an ESG or governance pillar score as a proxy for climate seriousness is not supported.

ESG-linked executive compensation specifically. Oyewo's positive coefficient is a direct challenge to the assumption that tying pay to ESG outcomes improves them. The plausible mechanism is that generic ESG-linked pay is easy to satisfy with peripheral action.

Headline net zero pledges. Dietz and Hastreiter's central result is that adoption is followed by little immediate change in either emissions or broad governance. Their framing is that such commitments are "neither purely symbolic nor immediately transformative, but are embedded within a gradual process of organisational change". Combined with Jiang, Kim and Lu on the absence of consequences for missing targets, and Bolton and Kacperczyk on selection, the presence of a pledge is close to uninformative on its own.

Single transition risk metrics. Fliegel's divergence result means any finding resting on one proxy is conditional on that proxy.

10. Design rules

Five rules follow from the review. They are addressed to anyone building firm-level empirical work or a predictive model in this area.

Construct the label before the features, and decompose it. The dependent variable should be a constant-perimeter, multi-year change in Scope 1 and 2 emissions with the Scope 2 accounting method fixed, computed on disclosed emissions for training. It should then be split into the part the firm can be credited with and the part it cannot. A fall in Scope 1 and 2 can come from energy intensity, fuel mix or electricity intensity, which are management, or from output, the grid emission factor and perimeter changes, which are not. Location-based Scope 2 is better than market-based here but still rewards operating in a country with a clean grid. Two firms can post near-identical headline reductions where one cut energy intensity by a sixth and the other merely shrank and sat in a decarbonising grid; a label that cannot tell them apart will train a model to identify firms in clean grids having bad years. Single-year changes carry too much noise, mixed Scope 2 methods measure procurement, and estimated emissions in the label mean the model partly learns the vendor's estimation function. Restatements from mergers, acquisitions and divestments are the largest practical hazard and have to be handled explicitly, which is what the chained construction is for.

Beat persistence before claiming a finding. Report every feature's contribution net of lagged emissions growth and sector by region fixed effects. A model that predicts intensity levels is answering a different question.

Weight indicators by rarity and re-estimate over time. Dietz and Hastreiter's weighted Management Quality result generalises. An indicator's information content is a function of how many firms satisfy it, which changes year by year. A feature set fixed in 2019 will quietly decay. Discriminatory power should be measured per indicator per period, and the decay itself is a finding worth reporting.

Keep red flags as a profile. Given phi correlations near zero and several negative pairs, summing the seven dimensions into a greenwashing score discards most of the information. Model them jointly, look at which combinations co-occur, and test whether particular combinations predict subsequent emissions differently.

Report a risk difference alongside any AUC. The two answer different questions and diverge systematically. AUC is rank-based and invariant to the base rate, so in a cell where few firms abate the handful that do are strongly selected and the ranking looks excellent while the indicator changes almost no outcomes. Simulated across the abatement gap, AUC reads 0.715 where the risk difference is 0.038. A study reporting AUC alone concludes an indicator works best in hard-to-abate sectors, which is the opposite of where it decides anything.

Treat sector by region as a mechanism, not a nuisance. Section 6 derives the inequality; the cell stands for its three terms: the effective carbon rate actually faced, the cost of the cheapest scaled abatement option, and the

premium buyers will pay for the low-carbon product. Abatement happens where all three clear together, which is why Ruiz Manuel and Blok find 86% of member reductions in eight electricity and heavy-industry firms, and why the same sector behaves differently across regions: European green steel premiums run EUR120-180 a tonne against a Chinese willingness to pay capped near a seventh of the cost gap. A fixed effect absorbs all of this, which is right for identification and leaves nothing to interpret and no way to extrapolate to a cell the sample never contained. A separate note, `SECTOR_REGION_MECHANISMS.md`, works through the evidence and the test for whether measured mechanisms can replace the dummies; `ABATEMENT_MECHANISMS.md` derives why the three behave as they do, from the condition under which a firm abates at all.

Vary the transition risk proxy and report the spread. Fliegel's recommendation is the right minimum standard. Where a result holds under taxonomy alignment, emission intensity and text-based measures alike, it is a result. Where it holds under one, it is a description of that measure.

11. What the field still needs

Four things, in rough order of how much difference they would make.

A location-based Scope 2 series with coverage comparable to market-based would settle the question this review is built around. At present the cleanest evidence comes from the subset of firms that dual-report, which is a selected sample.

Material-category Scope 3 disclosure, sector by sector, would make value chain emissions analysable. Volume of Scope 3 disclosure has risen without this happening.

Outcome tracking for targets, in the sense of a public record of what happened to every expiring target, would change the incentive structure Jiang, Kim and Lu document. The finding that 31% of targets disappear without comment is a data infrastructure failure as much as a corporate governance one.

Replication of the text-based indicators on reported rather than estimated emissions, or on a jurisdiction with mandatory assurance, would establish whether the cheap talk relationship is a behavioural fact or a property of an estimation model.

A note on what this review does not establish

Most of the studies here are observational. Colmer et al. is the exception with a credible causal design, and it addresses regulation rather than voluntary action. Schüder and Zülch use matching and a stacked difference-in-differences robustness check, and Dietz and Hastreiter use staggered difference-in-differences with matching, but both are working with voluntary adoption, where selection is severe and the authors say so. Silvia et al. is explicitly interpreted as associational.

The recurring pattern I have described, where improvement concentrates in procurement-sensitive and composition-sensitive measures, is consistent across enough independent designs that I am fairly confident in it. The firm-level predictors in Section 7 are on weaker ground, and the honest summary is that regulatory exposure and committed capital have reasonable support, rarity-weighted management practice has one good study behind it, and everything else is contested.

A note on the estimators

Because almost every study reviewed here rests on the same two techniques, it is worth being explicit about what they do and where they fail. Both are implemented in the accompanying code.

Matching. Adoption of a climate target is voluntary and strongly selected. Bolton and Kacperczyk find that committers already have lower emissions and that firms are less likely to make an ambitious commitment when their Scope 1 emissions are higher. Coarsened exact matching, as Schüder and Zülch use it, bins covariates, keeps only strata containing both adopters and non-adopters, and discards the rest. The discard is the cost and should always be reported: over-specify the covariate list and half the sample can disappear, at which point the estimate applies to a population that no longer resembles the one of interest.

Staggered difference-in-differences. Firms adopt in different years, so there is no single before and after. The standard two-way fixed effects specification handles this badly: with heterogeneous effects it uses already-treated firms as controls for later-treated ones, and the resulting weighted average can carry the wrong sign even when every underlying effect shares a sign. The alternative is to estimate a separate effect for each adoption cohort and period against a clean control group, either never-treated firms or those not yet treated, then aggregate by event time. Dietz and Hastreiter use this approach.

Two practical points. Standard errors should be clustered or bootstrapped at the firm level, since emissions are highly persistent within a firm and treating firm-years as independent shrinks standard errors by roughly the square root of the panel length. And pre-treatment estimates are part of the result, not a diagnostic to be dropped: Dietz and Hastreiter's significant coefficient in the year before adoption is what tells them firms begin acting before they announce.

Code and figures

Every analysis in this review is implemented in `backend/arp/decarb/`, with the econometrics in `backend/arp/decarb/research/`. `python -m arp.decarb.pipeline` reproduces the descriptive results and `python -m arp.decarb.research.figures` regenerates every figure.

All figures in this paper are generated from a synthetic panel, calibrated so that its headline statistics match the magnitudes reported in the literature reviewed here. They illustrate what each method does and what its output looks like. They are not empirical estimates, and no number in them is evidence about any company. The empirical claims in the text are sourced to the studies cited, not to the figures.

The synthetic panel exists because the underlying data is licensed. Running the same code against a real panel requires only building the input frame; the analysis functions take the same types either way.

Method extractions for all fifteen studies - sample construction, variable definitions, estimators, inference, target results and the specific obstacles to replicating each - are in [REPLICATION_SPECIFICATIONS.md](#).

The code implements estimator families, not replications. No number it produces reproduces a published result, and several methods central to the papers reviewed here are not implemented at all, among them the ClimateBert cheap-talk index, the ClimateBERT-NetZero classifier, Fliegel's return- sensitivity evaluation, and the EU ETS administrative-data design. A table in `backend/arp/decarb/README.md` sets out, paper by paper,

what is implemented and to what fidelity. One point of detail is worth recording here: LSEG state their attribution method in prose and publish the equation as an image, so the implementation was written from the prose. Two readings of that prose turn out to be algebraically identical except where a constituent's factors exactly cancel, which is the one case their footnote 39 pins down, and both are implemented.

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